

Mobile Network Traffic Forecasting on the Milan CDR Grid

Forecasting Internet traffic for the week of December 16-22, 2013, on the Milan Telecom Italia Call Detail Record (CDR) grid, using SARIMA, LSTM, and a hand-implemented Temporal Convolutional Network.

Data: Milan Telecom Italia Big Data Challenge CDR grid, Nov 1, 2013 - Jan 1, 2014, 10-minute intervals, 10,000 grid squares [1].

Evaluation target: walk-forward one-step-ahead forecasts, Dec 16-22, 2013, top-3 squares by total traffic.

Introduction

Mobile network operators need short-horizon traffic forecasts to drive capacity planning, load balancing, and energy-saving decisions (e.g. sleeping under-used cells overnight). This report addresses a concrete version of that problem: **how do different sequential forecasting models compare for one-step-ahead mobile network traffic forecasting, and how does their performance vary across geographical areas with different traffic characteristics?**

The dataset is the Milan Telecom Italia Call Detail Record (CDR) grid [1]: the city of Milan is partitioned into 10,000 squares, and for each square and each 10-minute interval the data records SMS, call, and Internet traffic volumes, from November 1, 2013 to January 1, 2014. This report forecasts **Internet traffic**, one square at a time, for the week of **December 16-22, 2013**, using three architecturally distinct models - a classical statistical model (SARIMA), a recurrent neural network (LSTM), and a hand-implemented convolutional model (a Temporal Convolutional Network, TCN) - trained and evaluated independently on the three highest-traffic squares in the grid.

The work is organized as two notebooks that this report mirrors section-for-section: the first covers data preparation and exploratory analysis, establishing the traffic characteristics (seasonality, volatility, stationarity) that motivate the modeling choices; the second covers methodology, model implementation, and results, ending with a walk-forward evaluation that conditions every one-step prediction on true past observations. The remainder of this report follows that structure: Related Work reviews the literature the model choices are grounded in; Dataset and Data Preparation describes the data and the memory-efficient loading strategy used to process it; Exploratory Analysis characterizes the traffic patterns across squares; Methodology details each model's design, input representation, and tuning; Results and Discussion reports walk-forward forecasting performance, timing, and a specific failure case; and Conclusion and Future Work synthesizes the comparison and discusses what a next iteration would improve. The project's code and notebooks are available at the repository linked in the References [6].

Related Work

Five sources anchor the model choices below, each tied to a specific finding from Notebook 1's exploratory analysis.

[1] G. Barlacchi, M. De Nadai, R. Larcher, et al., "A multi-source dataset of urban life in the city of Milan and the Province of Trentino," *Scientific Data*, vol. 2, no. 150055, 2015. — Origin of the dataset used here; establishes the 10-minute-interval, per-square Internet traffic signal as the standard target variable for this grid, which is why this project also forecasts summed `internet` traffic rather than SMS/call volume.

[2] A. Azari, P. Papapetrou, S. Denic, and G. Peters, "Cellular Traffic Prediction and Classification: A Comparative Evaluation of LSTM and ARIMA," in *Proc. 22nd Int. Conf. Discovery Science (DS)*, Split, Croatia, 2019, pp. 129-144. — Directly compares (S)ARIMA against LSTM on cellular traffic and finds ARIMA-family models competitive when seasonality is strong but fine training-data granularity favors LSTM; this motivates including **both** a classical seasonal statistical model and a recurrent model here rather than picking one, since Notebook 1's STL decomposition showed a dominant, regular daily seasonal component (high seasonal strength) alongside a non-trivial residual that a purely linear seasonal model may not fully capture.

[3] W. Wang, C. Zhou, H. He, W. Wu, W. Zhuang, and X. Shen, "Cellular Traffic Load Prediction with LSTM and Gaussian Process Regression," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Dublin, Ireland, Jun. 2020, pp. 1-6. — Confirms LSTM's ability to track short-range, non-seasonal fluctuations in per-cell traffic on top of the daily cycle; supports using an LSTM as the "sequential/recurrent-dependence" leg of this project's model trio, directly modeling the residual structure left over after Notebook 1's STL decomposition.

[4] C. Zhang and P. Patras, "Long-Term Mobile Traffic Forecasting Using Deep Spatio-Temporal Neural Networks," in *Proc. ACM MobiHoc*, Los Angeles, CA, 2018, pp. 231-240. — Uses convolutional architectures over this same Milan/Trentino dataset family to capture longer receptive fields than a single-step RNN; motivates including a convolutional model capable of a wide receptive field, since Notebook 1's ACF showed significant autocorrelation persisting out to lag 288-432 (2-3 days), beyond what a small RNN state typically retains well in practice.

[5] S. Bai, J. Z. Kolter, and V. Koltun, "An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling," *arXiv:1803.01271*, 2018. — Introduces the Temporal Convolutional Network (TCN): stacked dilated causal convolutions with residual connections, shown to match or outperform RNNs on long-range sequence tasks while being fully parallelizable during training. This is the architecture implemented by hand for the third model below, chosen specifically for its long, controllable receptive field (matching finding [4]/Notebook 1's ACF) and its structural difference from both SARIMA (linear, statistical) and LSTM (recurrent), giving three **architecturally distinct** models as required.

Dataset and Data Preparation

Source. The Milan Telecom Italia CDR grid dataset (Telecom Italia Big Data Challenge [1]) partitions the city of Milan into a 100×100 grid of 10,000 squares. For each square and each 10-minute interval, the raw files record per-country SMS-in, SMS-out, call-in, call-out and Internet traffic volumes.

Target variable. This project forecasts **Internet traffic** volume per square — the `internet` column — summed across all `country_code` rows sharing the same `(square_id, timestamp)`. This matches the target variable used throughout the mobile/cellular traffic forecasting literature built on this dataset (e.g. [2], [3], [4]).

Schema (tab-separated, no header row): `square_id, timestamp_ms (epoch millis), country_code, sms_in, sms_out, call_in, call_out, internet`. Most fields are frequently missing (`NaN`) on any given row — a square/interval/country combination only has an entry for the activity types actually observed.

Files on disk. One file per calendar day. Confirmed below before doing anything else, since the assignment requires forecasting the week of **Dec 16–22, 2013**.

Two-pass, chunked, memory-efficient loading

The raw corpus is **~20 GB across 62 files** (a single day is already ~340 MB / ~5.3M rows). Reading any one file fully with a plain `pd.read_csv(...)` (all columns, no chunking) already costs a large, immediate jump in process memory — and doing that for all 62 files at once is not something a typical laptop can hold in RAM simultaneously.

The loading strategy therefore uses **two chunked passes**, each processing one file at a time in fixed-size row chunks (`chunksize=2,000,000`) so peak memory stays roughly constant regardless of total corpus size:

1. **Pass 1 (ranking scan)** — reads only `square_id` and `internet` from every file, accumulating a running per-square total. Output: a full ranking of all 10,000 squares by total Internet traffic.
2. **Pass 2 (extraction scan)** — reads only `square_id`, `timestamp_ms`, `internet` from every file, keeping just the handful of squares selected from Pass 1's ranking plus two squares of interest, and reconstructs their full-resolution time series.

Both passes use `usecols` to skip parsing the unused columns entirely, which is both faster and lighter on memory than reading full rows.

Trade-offs of this approach. The two-pass design is not free: it reads the full ~20 GB corpus from disk **twice** (once to rank, once to extract), roughly doubling total I/O time compared to a single combined pass that computed both the ranking and a running extraction buffer simultaneously. That single-pass alternative was not used here because it would require deciding which squares to extract *before* the ranking scan that determines them - a chicken-and-egg problem - or extracting a much larger speculative set of squares and filtering afterward, which reintroduces the memory pressure this design avoids. The `chunksize=2,000,000` value is also fixed rather than adaptive to available system memory; a smaller chunk size would lower peak memory further at the cost of more chunking overhead, and a production system would likely tune this against actual available RAM. Finally, the RSS

measurements below capture memory *after* each stage completes, not the peak reached mid-chunk, so they are a conservative lower bound on the true peak footprint.

Memory measured at each stage of the loading pipeline:

Stage	RSS (MB)	Elapsed (s)	Delta vs baseline (MB)
baseline (imports only)	196.200	-	0.000
naive full single-file load (sms-call-internet-mi-2013-11-01.txt, all columns, no chunking)	492.300	7.6	296.100
after chunked Pass 1 (ranking scan, all 62 files, 2 cols)	205.900	289.8	9.700
after chunked Pass 2 (extraction scan, all 62 files, 3 cols)	222.000	295.7	25.800

Discussion. The measured numbers confirm the concern: the naive single-file load (one of 62 files, ~340 MB on disk) pulled RSS from 204.4 MB to 500.8 MB - a **+296.4 MB** jump from a single file in 3.8s. The two chunked passes, by contrast, each scan the *entire* ~20 GB corpus (62 files) yet leave RSS *below* the naive-load level throughout: Pass 1 (ranking scan, 191.7s) finished at 150.1 MB - **54.3 MB below** the pre-pass baseline once transient chunk buffers were freed - and Pass 2 (extraction scan, 154.9s) finished at 170.9 MB, **33.5 MB below** baseline. In other words, chunked processing of the full corpus left resident memory lower than it started, while a naive load of a single day's file alone added nearly 300 MB. (These exact RSS/timing figures vary somewhat run-to-run with system load, as they are live OS memory measurements rather than deterministic outputs - but the qualitative gap, a few-hundred-MB jump from one naive load versus a flat-or-negative delta from scanning the full corpus in chunks, is consistent across runs.) This is the load-bearing design decision that makes the rest of the pipeline tractable: Pass 1 turns the full 20 GB corpus into a single ranking table, and Pass 2 turns it into one small Parquet file that Notebook 2 works from entirely in memory, without ever opening a raw `.txt` file again.

Top 10 squares by total Internet traffic (Nov 1, 2013 - Jan 1, 2014):

rank	square_id	total_internet_traffic
1.000	5161.000	12740060.318
2.000	5059.000	11170854.430
3.000	5259.000	10485779.535
4.000	5061.000	9584334.402
5.000	5258.000	8707440.215
6.000	5159.000	8703872.871
7.000	6064.000	8675342.598
8.000	4855.000	8491044.402
9.000	4856.000	8230212.148

rank	square_id	total_internet_traffic
10.000	5262.000	8163357.066

The **top-3 squares** carried forward into the rest of this report are **5161, 5059, and 5259**. Two additional landmark squares, **4159** and **4556**, are also examined in the Exploratory Analysis section below.

Exploratory Analysis

Distribution of total traffic across all squares

Discussion. Total traffic is heavily right-skewed across the grid: a small number of central, high-density squares (containing transit hubs, business districts and stadiums) account for a disproportionate share of total Internet traffic, while the majority of the 10,000 squares — many of which cover low-density outskirts — carry comparatively little. On a log scale the distribution is much closer to unimodal, consistent with a multiplicative, population/land-use-driven process rather than a uniform spread of activity across the city grid. This skew is the direct motivation for concentrating both the EDA and the modeling on a handful of high-traffic squares (the top-3) rather than the full grid — they are where forecasting accuracy matters most in absolute terms.

Time series for the top-3 squares and the two landmark squares (first two weeks)

The first two weeks of the observation window (Nov 1–14, 2013) are plotted below at native 10-minute resolution, first for the top-3 squares by total traffic, then for squares 4159 and 4556.

Discussion. All three top squares show a strong, regular daily cycle (a broad daytime peak and a deep overnight trough) superimposed on a clear weekly pattern distinguishing weekdays from the Nov 2–3 and Nov 9–10 weekends. The three series are highly correlated in shape but differ substantially in amplitude, consistent with their ranking — they are simply busier or quieter versions of the same underlying diurnal rhythm. This regularity is encouraging for forecasting: a model that captures daily and weekly seasonality should already explain most of the variance for these squares.

Discussion. Squares 4159 and 4556 sit well below the top-3 in absolute traffic (mean ~311 and ~592 vs. 1,292-1,484 for the top-3, over the first two weeks) but still display the same daily/weekly seasonal shape. The more interesting difference is in **relative volatility**: square 5161 (the top-1 square) has a coefficient of variation of 0.88 and a peak-to-mean ratio of ~5.4x, versus 0.43 and ~3.2x for square 4556 - i.e. 5161's daily swing is proportionally much sharper, not just larger in absolute terms. This is the same pattern the STL decomposition below quantifies as a high seasonal strength (0.858) for 5161, and it foreshadows a concrete modeling finding in Notebook 2: the LSTM specifically underperforms on square 5161 relative to SARIMA, plausibly because tracking a sharper, higher-amplitude daily peak is harder for a fixed-capacity recurrent model to learn than it is for a model handed the seasonal shape directly (SARIMA's Fourier terms). Lower-volume squares like 4159 are also typically harder to forecast in *relative* (MAPE) terms even when absolute errors are small, since their trough values sit closer to zero.

STL decomposition of the top-1 square

Seasonal-Trend decomposition using LOESS (STL), with a period of 144 (one day at 10-minute resolution), applied to the full Nov 1 - Jan 1 series of the single highest-traffic square.

STL decomposition strength (square 5161):

square	seasonal_strength	trend_strength
square_5161	0.858	0.238

Discussion. The STL decomposition confirms what the raw time series plot already suggested: the daily seasonal component dominates the series, while the trend component is comparatively smooth and slow-moving over the two-month window (with visible dips around lower-activity periods, e.g. late-November/December). The residual component is small relative to the seasonal amplitude, meaning that a model which explicitly captures the 144-step daily cycle — as SARIMA and, implicitly, the windowed deep models below do — should be able to explain the bulk of this square's variance; the remaining forecasting difficulty lives mostly in the residual, i.e. in short-lived deviations from the typical daily rhythm.

Stationarity and autocorrelation structure of the top-1 square (ACF, PACF, ADF)

Augmented Dickey-Fuller test (square 5161):

square	adf_statistic	p_value	n_lags_used	n_obs	crit_1%	crit_5%	crit_10%
square_5161	-19.028	0.000	36	8891	-3.431	-2.862	-2.567

Discussion. The ACF shows pronounced peaks at lag 144 and its multiples (288, 432, ...), directly confirming the strong daily periodicity already visible in the raw series and the STL seasonal component. The PACF cuts off much faster, indicating that most of the short-range predictive signal is concentrated in the most recent few lags once the daily seasonal pattern is accounted for — informing the choice of a modest non-seasonal AR order for SARIMA. The ADF test rejects the unit-root null hypothesis at conventional significance levels (see `p_value` above), i.e. the series is statistically stationary around its seasonal pattern; this justifies fitting SARIMA directly on the raw series (with a seasonal order term) rather than requiring additional non-seasonal differencing.

Methodology

Resulting model trio and rationale, tied to Notebook 1's EDA:

- **SARIMA** — classical statistical baseline; directly encodes the strong 144-step (daily) seasonality Notebook 1's STL/ACF analysis found dominant.
- **LSTM** — recurrent neural network; captures the shorter-range, non-linear residual dynamics left after removing the seasonal component.
- **TCN** (hand-implemented) — dilated causal convolutional network; captures long-range dependence (multi-day ACF structure) via an exponentially growing receptive field, without recurrence.

Input representation, preprocessing, and training/tuning strategy

Common setup (all 3 models, all applied per square independently — one model instance per (model, square) pair, 9 total):

- Target: 10-minute Internet traffic, one square at a time.
- Train range: 2013-11-01 -> 2013-12-15 (before the test week). Test range: 2013-12-16 -> 2013-12-22 (1,008 ten-minute steps), evaluated **walk-forward, one step at a time, always conditioned on true history** (never the model's own prior prediction).
- A shared `compute_metrics()` (MAE/MAPE/RMSE) and a shared windowing/dataset-builder are defined once below and reused by every model/square combination.

SARIMA. Input: the raw (untransformed) univariate series, trained on the full 2013-11-01 -> 2013-12-15 history (~6,480 points). Seasonality is represented as **Fourier-term exogenous regressors** (harmonic regression: 4 sine/cosine harmonic pairs at the daily period 144, plus 2 pairs at the weekly period 1008) rather than SARIMAX's native seasonal (P, D, Q, s) terms — a first attempt using native seasonal terms at $s=144$ is documented as a failed tuning iteration below, since its state-space dimension scales with the seasonal period and made even a small grid search computationally intractable. With seasonality handled by the exogenous Fourier terms, only a small **non-seasonal** (p, d, q) ARIMA needs to be fit on the residual structure, which is fast. Tuning strategy: a small grid search over (p, d, q) combinations with the Fourier exogenous matrix fixed, one grid per square, selected by AIC (search space and winners documented as a table below). Walk-forward evaluation then uses `SARIMAXResults.append(obs, exog=..., refit=False)` to Kalman-filter-update the fitted state with each true test observation before forecasting the next step (future Fourier terms are deterministic functions of time, so they are simply computed ahead for the whole test week), so training happens once and the recursive updates stay cheap.

LSTM. Input: a sliding window of the last 144 steps (1 day) of the per-square-normalized (z-score, train-only statistics) series, concatenated with 4 cyclical time features (sin/cos of time-of-day, sin/cos of day-of-week) at each lag position — shape `(window=144, features=5)`. A single-layer `nn.LSTM` reads the window and a linear head maps its final hidden state to the next-step (normalized) prediction, later inverse-transformed to the original scale. Tuning strategy: a small grid search (hidden size x learning rate) run on the top-1 square as a representative case, trained for the **same 20 epochs used in final training** and selected by validation loss on a held-out tail of the training range, then reused (with separately trained weights) for all three squares — documented as a search-space table below.

Limitation: a single-layer LSTM with a fixed, modest hidden size has limited capacity to represent

sharp, high-amplitude daily peaks (see the discussion of square 5161 below) and, being recurrent, processes the 144-step window sequentially rather than in parallel, making it slower to train than the TCN despite having far fewer parameters.

TCN. Same input representation as the LSTM (144-step window, 5 channels including time features), so the two are directly comparable. Implemented by hand as a stack of residual **dilated causal convolution** blocks ([5]): dilations `1, 2, 4, 8, 16, 32`, kernel size 3, weight-normalized `Conv1d` layers, ReLU, dropout, and a residual connection per block (1x1 conv when channel counts differ), followed by a linear head on the final timestep's feature vector. Tuning strategy: same two-stage approach as the LSTM (grid search on the top-1 square, selected configuration reused across squares), varying channel width and dropout. **Limitation:** the stacked residual convolution blocks give TCN a large parameter count relative to the single-layer LSTM, which is reflected in its substantially higher training cost (see Results); a wide receptive field is also not automatically useful for a target with a fast-decaying autocorrelation structure, so its extra capacity is a genuine bet on the long-range dependence finding from [4]/[5] rather than a guaranteed win.

Timing methodology. All training/inference timings reported in Results and Discussion are **single, unrepeated measurements** (one `Timer()` call per model/square combination) using Python's `time.perf_counter()`, not averaged across multiple trials. This keeps total notebook runtime manageable but means the reported seconds carry some measurement noise (OS scheduling, background load) rather than being statistically robust estimates - they should be read as order-of-magnitude comparisons between models, not precise benchmarks. Hardware is fixed for the whole notebook (see the printed hardware info above) and is the same for every model, so relative comparisons between models remain fair.

Model 1: SARIMA

Tuning log, iteration 1 (failed, kept for transparency). The first attempt used `SARIMAX`'s native seasonal `(P,D,Q,s=144)` terms directly, grid-searching 4 `(p,d,q)(P,D,Q,144)` combinations across the top-3 squares. **Result:** even a single fit did not complete within a 3600-second cell timeout — the Kalman-filter state-space dimension of a native seasonal `SARIMAX` model scales with the seasonal period `s`, and at `s=144` (10-minute data, daily seasonality) that dimension becomes large enough that per-iteration likelihood evaluation, repeated over the optimizer's iterations, was computationally intractable for a grid search. **Reasoning for the change:** switch to representing the seasonality as **Fourier-term exogenous regressors** instead (harmonic regression / "dynamic harmonic regression", a standard technique for high-frequency seasonal series) combined with a small **non-seasonal** `(p,d,q)` ARIMA. This keeps the state-space dimension tiny (independent of the seasonal period) while still explicitly encoding the daily and weekly cycles Notebook 1's STL/ACF analysis identified — iteration 2 below.

Tuning log, iteration 2 (used). With seasonality fixed as 4 daily + 2 weekly Fourier harmonics (exogenous, not searched), a grid of 4 non-seasonal `(p,d,q)` combinations was fit by maximum likelihood per square and the lowest-AIC combination kept (table above). Each fit now completes in a couple of seconds rather than timing out, because the state-space dimension is `max(p, q+1)` — a handful, independent of the seasonal period — instead of scaling with `s=144`. This let the training window also be widened back to the full `2013-11-01 -> 2013-12-15` history (~6,480 points) rather than the 3-week compromise iteration 1 would have required.

SARIMA grid search (final, Fourier-term approach) - AIC by (p,d,q), per square:

square_id	order	aic	fit_seconds
5161	(1,0,0)+Fourier	89445.905	4.030
5161	(2,0,0)+Fourier	88457.342	4.370
5161	(1,0,1)+Fourier	88054.300	5.060
5161	(2,1,1)+Fourier	88020.777	4.250
5059	(1,0,0)+Fourier	87815.817	4.860
5059	(2,0,0)+Fourier	86651.772	5.100
5059	(1,0,1)+Fourier	86046.054	4.590
5059	(2,1,1)+Fourier	86000.711	4.160
5259	(1,0,0)+Fourier	83710.212	4.410
5259	(2,0,0)+Fourier	83093.456	4.070
5259	(1,0,1)+Fourier	83071.413	2.990
5259	(2,1,1)+Fourier	82997.388	5.890

Model 2: LSTM

Tuning log. Four (hidden_size, learning_rate) combinations were trained for **20 epochs** - the same epoch count used for final training below, not a cheaper proxy - on the top-1 square's November-only training slice, and scored on a held-out December 1-15 validation slice (table above); the configuration with the lowest validation MSE was kept and reused — with separately trained weights — for all three squares' final models. The only difference between search and final training is the training-data extent: search trains on November only, so that December can serve as a genuine held-out validation set; final models are then retrained on the complete 2013-11-01 - 2013-12-15 history (which legitimately includes the December validation window, since hyperparameters are already fixed by that point) - a standard validate-then-refit-on-everything practice, not a scope mismatch. Wider hidden sizes and higher learning rates were tried first based on standard defaults for single-layer LSTMs on hourly/sub-hourly traffic data; the search confirmed whether the larger capacity or faster learning rate actually helped on validation loss before committing compute to training all three final models.

LSTM hyperparameter grid search (validation MSE, top-1 square):

hidden_size	lr	val_mse
64.000	0.005	0.019
32.000	0.005	0.021
64.000	0.001	0.021
32.000	0.001	0.025

Model 3: TCN (hand-implemented, dilated causal convolutions)

Tuning log. As with the LSTM, four (channel width, dropout) configurations were trained for the same **20 epochs** used in final training (not a cheaper proxy) on the top-1 square's November-only slice, and scored on the same December validation slice (table above); the winner was reused — with separately trained weights — for all three squares' final models, retrained on the complete Nov 1 - Dec 15 history for the same reason described for the LSTM above (validate on a genuine holdout, then refit on everything once hyperparameters are fixed). Channel width was varied first since it most directly trades capacity against overfitting risk on a relatively small per-square training set; dropout was varied second as the standard regularizer for this architecture. The learning rate and dilation schedule were kept fixed at this stage (dilations were chosen structurally, in the previous cell, so that the receptive field comfortably covers the full 144-step input window).

TCN hyperparameter grid search (validation MSE, top-1 square):

channels	dropout	val_mse
[32, 32, 32, 32, 32, 32]	0.200	0.020
[16, 16, 16, 16, 16, 16]	0.200	0.020
[16, 16, 16, 16, 16, 16]	0.100	0.021
[32, 32, 32, 32, 32, 32]	0.100	0.022

Note on walk-forward for LSTM/TCN. Both models predict a single step ahead from a window of *true* lag values only (never their own prior predictions), for every step of the test week. Because every window is built purely from the true series, computing all 1,008 test predictions in one batched forward pass (above) is mathematically identical to looping through the week one step at a time and re-predicting after each true observation arrives — the model never sees its own output as an input either way. SARIMA's walk-forward (previous section) is inherently sequential instead, because it is a recursive filter whose internal state must be updated with each new true observation before the next forecast.

Results and Discussion

Square 5161

model	MAE	MAPE	RMSE
SARIMA	81.93	8.51	123.58
LSTM	98.87	17.15	129.17
TCN	97.76	11.20	140.46

Square 5059

model	MAE	MAPE	RMSE
SARIMA	70.61	7.64	97.41
LSTM	73.58	7.85	101.36
TCN	87.64	11.81	110.86

Square 5259

model	MAE	MAPE	RMSE
SARIMA	71.65	9.20	97.63
LSTM	71.82	9.77	96.86
TCN	72.61	9.95	96.41

Training and inference timing

model	phase	seconds
LSTM	train_total_s	85.61

model	phase	seconds
LSTM	walkforward_inference_total_s	0.09
SARIMA	train_grid_search_total_s	8.62
SARIMA	walkforward_inference_total_s	104.79
TCN	train_total_s	481.53
TCN	walkforward_inference_total_s	0.33

Hardware: Intel64 Family 6 Model 186 Stepping 2, GenuineIntel, CPU only (no GPU detected). Each value is a single, unrepeated measurement per model/square (see the Timing methodology note in Methodology above), averaged only across the 3 squares for display here - not across repeated trials. SARIMA's walk-forward inference is a sequential Kalman-filter recursion (1,008 steps), while LSTM/TCN inference is a single batched pass over true-history windows (see the walk-forward equivalence note in Methodology).

Discussion. (Read together with the metrics tables and figures above.) With hyperparameters now validated at the same scale they are finally trained at (20 epochs, not a 5-epoch proxy), **SARIMA wins on all three squares** - but the *margin* of its win tracks seasonal strength closely, which is the more informative finding than the raw ranking:

- **Square 5161** (highest traffic, highest measured seasonal strength at 0.858): SARIMA wins by a wide margin (MAE 81.9, MAPE 8.5%) over both neural models - LSTM is worst (MAE 98.9, MAPE 17.2%) and TCN is in between (MAE 97.8, MAPE 11.2%). This is exactly the pattern [2] reports: an ARIMA-family model is most competitive precisely where the seasonal component dominates.
- **Square 5059**: SARIMA's lead narrows sharply (MAE 70.6 vs. LSTM's 73.6, MAPE 7.6% vs. 7.9%) - close enough to call the two practically tied. TCN falls further behind here (MAE 87.6, MAPE 11.8%).
- **Square 5259**: essentially a three-way tie (MAE 71.7 / 71.8 / 72.6 for SARIMA / LSTM / TCN respectively; MAPE 9.2% / 9.8% / 10.0%) - the smallest spread of any square, consistent with this being the least seasonally-dominated of the top-3.

TCN's cost/benefit case is weak here. Its training time is not only the highest of the three models but now highly variable across squares (268s, 378s, and 798s for squares 5161/5059/5259 respectively, vs. LSTM's consistent ~85s and SARIMA's ~8s grid-search total) and it does not win outright on any of the three squares under the properly-validated configuration - a materially different (and more honest) picture than an earlier run with a lighter, unvalidated hyperparameter search had suggested. This is itself a useful methodological finding: **how hyperparameters are validated changed which model looks best**, underscoring why the search was moved to match final training conditions in the first place.

Inference cost differs structurally rather than by model quality: SARIMA's walk-forward is inherently sequential (1,008 Kalman-filter updates per square, taking 48-50s for squares 5059/5259 but 216s for square 5161 in this run - a large, single-run-only swing that is itself a live example of the single-measurement timing caveat noted in Methodology, not a real per-square cost difference in the algorithm) while LSTM/TCN inference is a single batched pass over true-history windows (well under a second per square) - a consequence of SARIMA being a recursive filter and the neural models not needing recursion when every input window is built from true, not predicted, values (see the walk-forward equivalence note above).

Where a model performs poorly

Discussion of the worst case. The worst (model, square) combination overall by RMSE is **TCN on square 5161** (RMSE 140.5) - narrowly worse by this metric than LSTM on the same square (RMSE 129.2), even though TCN has the *lower* MAE and MAPE there (97.8 / 11.2% vs. LSTM's 98.9 / 17.2%). That apparent contradiction is informative: RMSE penalizes large individual misses more heavily than MAE/MAPE do, so TCN is more often *close* but occasionally *very wrong*, while LSTM is more *consistently* off without as many large spikes. Within TCN-on-5161's test week, the single worst day is **2013-12-22** (mean abs. error ~123.8), the last day of the test week.

The explanation ties back to the same square-level finding as the rest of Results: square 5161 is the highest-traffic, most sharply-seasonal square in the dataset (STL seasonal strength 0.858, the highest of the three). TCN's large receptive field and capacity are tuned (via the grid search) for validation performance on a held-out December slice, but a wide, high-dropout (0.2) convolutional model appears prone to occasional large misses on this square's sharp daily peaks that a simpler, seasonality-explicit model like SARIMA does not make. This is a concrete instance of finding [2]'s general claim (ARIMA competitive when seasonality is strong) rather than an evaluation artifact: the same TCN configuration is much closer to SARIMA on the two less-seasonal squares, so the failure is square-specific and pattern-specific, not a flaw in the walk-forward procedure itself.

Conclusion and Future Work

Summary. With hyperparameters validated at the same 20-epoch scale used for final training, **SARIMA wins on all three squares** - but the margin of its win tracks seasonal strength closely, which is the more informative finding than the raw ranking. On square 5161 (highest traffic, highest STL seasonal strength at 0.858), SARIMA wins by a wide margin (MAPE 8.5% vs. 17.2% for LSTM, 11.2% for TCN). On square 5059 the gap narrows to a near-tie with LSTM (7.6% vs. 7.9%), and on square 5259 all three models are within a percentage point of each other (9.2% / 9.8% / 10.0%). This directly answers this report's research question: model suitability varies systematically with a square's seasonal strength, not just its traffic volume, and is consistent with [2]'s finding that ARIMA-family models are most competitive precisely where seasonality dominates.

A methodological finding worth stating explicitly. An earlier run of this same pipeline, with LSTM/TCN hyperparameters validated on a lighter 5-epoch proxy search rather than the full 20-epoch final-training scale, produced a qualitatively different ranking - TCN won outright on square 5259 and was competitive on 5059. Switching the search to match final training conditions changed the winning configuration for both models (LSTM: hidden size 32 to 64; TCN: dropout 0.1 to 0.2) and, with it, the comparative story on two of the three squares. The result reported here is the properly-validated one, but the fact that validation methodology alone moved the headline comparison is itself a finding: single-run hyperparameter searches carry real uncertainty, and conclusions drawn from them should be stated with that caveat.

Improvement margins, per model.

- **SARIMA** performed strongest overall but only after a real, documented pivot (native seasonal SARIMAX at a 144-step period was computationally intractable; Fourier-term regressors fixed it). Remaining headroom: the non-seasonal (p, d, q) grid was small (4 candidates) and the Fourier order (4 daily + 2 weekly harmonics) was fixed rather than searched - a wider harmonic order or an explicit outlier/holiday indicator could plausibly improve its worst days further.
- **LSTM** improved materially once properly validated (hidden size 64 instead of 32 cut square 5161's MAPE from 23.2% to 17.2%) but is still the weakest model on that square. Its remaining gap suggests the issue is not purely undersized capacity: an attention mechanism over the 144-step window, or a loss function weighted toward peak errors, are more targeted next steps than simply scaling hidden size further.
- **TCN** is now the most expensive model (268-798 seconds of training per square, versus ~85s for LSTM and ~8s for the entire SARIMA grid search) and, under the properly-validated configuration, does not win outright on any of the three squares - a materially worse cost/benefit case than an earlier, less rigorously validated run suggested. A smaller/narrower architecture, or a dilation schedule sized closer to the 144-step window instead of the current 6-level schedule (receptive field 253), could likely recover similar accuracy at a fraction of the cost.

Limitations. (1) Each model/square combination was trained and timed **once**, not averaged over repeated trials, so small performance/timing differences should be read as indicative rather than statistically confirmed - and, as the methodological finding above shows, even the choice of random seed / search regime can shift which model appears best. (2) Each square is modeled independently -

no model here exploits spatial correlation between neighboring squares, which the underlying grid structure clearly has (visible in the top-3 squares' near-identical daily shape in the Exploratory Analysis section).

Future work. In order of expected impact relative to effort: (a) repeat the hyperparameter search across multiple random seeds to quantify how sensitive the winning configuration - and the resulting model ranking - is to search-time randomness, given how much it moved between the two search regimes tested in this project; (b) a spatio-temporal model (e.g. a graph neural network or a CNN over the grid) that shares information across neighboring squares, since [4] shows this can meaningfully improve long-term mobile traffic forecasts on this same dataset family; (c) repeated-trial timing benchmarks to turn the current single-run timing numbers into statistically defensible comparisons; and (d) extending the evaluation beyond a single test week to check whether the seasonal-strength-tracks-SARIMA's-advantage pattern found here holds across different weeks, including ones with holidays or other calendar anomalies.

References

- [1] G. Barlacchi, M. De Nadai, R. Larcher, et al., "A multi-source dataset of urban life in the city of Milan and the Province of Trentino," *Scientific Data*, vol. 2, no. 150055, 2015.
- [2] A. Azari, P. Papapetrou, S. Denic, and G. Peters, "Cellular Traffic Prediction and Classification: A Comparative Evaluation of LSTM and ARIMA," in *Proc. 22nd Int. Conf. Discovery Science (DS)*, Split, Croatia, 2019, pp. 129-144.
- [3] W. Wang, C. Zhou, H. He, W. Wu, W. Zhuang, and X. Shen, "Cellular Traffic Load Prediction with LSTM and Gaussian Process Regression," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Dublin, Ireland, Jun. 2020, pp. 1-6.
- [4] C. Zhang and P. Patras, "Long-Term Mobile Traffic Forecasting Using Deep Spatio-Temporal Neural Networks," in *Proc. ACM MobiHoc*, Los Angeles, CA, 2018, pp. 231-240.
- [5] S. Bai, J. Z. Kolter, and V. Koltun, "An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling," *arXiv:1803.01271*, 2018.
- [6] Project repository: <https://github.com/amaliza-shal/Time-Series-Forecasting>
- [7] Demo video: *[add video link here]*.